

Validation Of A Novel Respiratory Monitoring Method and System Based on Antenna Sensors and Optical Tracking of Chest Motion

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Abstract—The effectiveness of a wearable sinusoidal dipole antenna sensor in monitoring respiratory activity is investigated. An experimental setup combining the antenna sensor and a highly accurate optical tracking system is utilized, and a strong correlation between the antenna's reflection signal and the optical tracker is demonstrated during various respiratory behaviors. Obtained results show the antenna sensor's high accuracy in detecting respiratory cycles, amplitudes, and identifying sleep apnea events. The maximum mean squared error in cycle timings is calculated as 0.0044 seconds squared, and the maximum relative error of measured amplitudes is calculated as $\pm 3.95\%$. Overall, these findings underscore the potential of sinusoidal antenna sensors as an effective tool for respiratory monitoring.

Keywords—Sinusoidal Antenna, Antenna Sensor, Wearable Antenna, Wearable Sensor, Respiration, Optical Tracking, Sleep Apnea, Hyperventilation, Hypoventilation, Hyperpnea, Hypopnea

I. INTRODUCTION

The respiratory rate is a critical vital sign, and its abnormalities are associated with diverse pathological conditions such as apnea, dyspnea, and pneumonia [1]. It is known as a significant predictor of cardiac events and severe medical conditions [2, 3], and even mental and physical conditions such as emotional stress and physical effort [4]. It is also an essential parameter for the overall health assessment of at-risk populations, such as older adults, newborns, post-surgery patients, and addicted individuals [5].

Many different methods have been explored to develop an accurate and reliable solution for monitoring respiratory activity. Some recent works such as Smart Facemask for cyclists [6], a low-cost Surface Acoustic Wave Sensor with a high sensitivity to airflow humidity [7], and a lightweight retractable thin-film sensor implemented in the form of a strap [8] are good examples for such methods. An emerging novel method for respiration monitoring is employing on-body antenna sensors, which have shown promising results, such as spiral-shaped fiber antenna [9], embroidered meander dipole antenna [10], and circular patch E-textile antenna [11].

Recently, a sinusoidal-shaped dipole antenna made of flexible conductors has been introduced for wearable applications with strain-sensing capabilities [12]. The present work leverages such an antenna to detect respiratory activity. During the respiratory cycle, the circumference of the thorax is naturally expanded, which can be identified as a sign of inspiration, followed by expiration, in which the circumference shrinks back to the initial size. This change in the circumference can be leveraged for detecting the respiratory cycle [13].

The high sensitivity of the sinusoidal antenna sensor to axial stretching makes it a potentially viable strain sensor for respiratory monitoring applications. The correlation between the sensor's response and the actual deformation of the antenna worn over the body needs to be studied to examine the performance of such antenna sensors for on-body strain sensing applications. This work focuses on validating the sensor's response by employing a highly accurate optical tracking system as the reference to confirm the accuracy of utilizing flexible sinusoidal dipole antennas in wearables to detect thorax expansion and, therefore, to detect respiratory activity. This paper is organized as follows: Chapter II describes the setup, Chapter III explains the experimental results, and Chapter IV provides discussions and the conclusion.

II. THE SYSTEM DESCRIPTION

A. Sensor Design and Configuration

The antenna sensor employed is a sinusoidal dipole antenna fabricated using conductive polymer wire designed for wearable applications. The conductive polymer is a composition of poly (ethylene-co-vinyl acetate) (PEVA) polymer and multi-walled carbon nanotubes (MWCNTs) [12]. For ease of use in the laboratory environment, the sensor is sewn over a flexible band, which can be worn on top of the subject's chest area. As shown in Fig. 1, the antenna is fabricated with a total axial length of $L = 80 \text{ mm}$ and a total peak-to-peak height of $H = 18 \text{ mm}$ with three peaks in each pole of the sinusoidal design, consuming a total of 142 mm polymer wire. The 50Ω antenna is designed for operation near the license-free 915 MHz ISM band when installed over the body. Fig. 2 represents the return

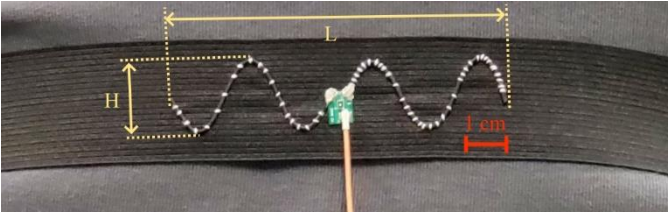


Fig. 1. Wearable Flexible Sinusoidal Dipole Antenna, sewn on a elastic band and worn around the thorax of a volunteer.

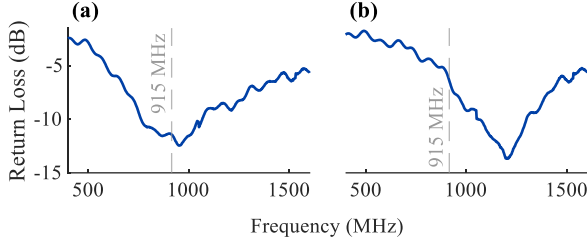


Fig. 2. The measured Return Loss of the fabricated antenna sensor (a) over the body and (b) in free space.

loss of the antenna sensor in free space and over the body. Table 1 summarizes the measured specifications of the antenna.

The antenna sensor is connected to a VNA system, and its return loss is continuously measured at a single frequency of 915 MHz. The measured phase and magnitude values are logged in a CSV file for the duration of the recording. The measurement runs every 20 ms, resulting in a sampling frequency of 50 Hz. Considering the respiratory frequency range of a healthy adult as 12 – 20 breaths per minute [14], the sampling rate is 150 times higher than the maximum respiration frequency, resulting in a clear waveform with minimal quantization noise.

TABLE I. MEASURED SPECIFICATIONS OF ANTENNA SENSOR

Configuration	Over the Body	In Free Space
Operation Frequency (f)	951 MHz	1190 MHz
Return loss at f	-12.46 dB	-13.41 dB
Impedance at f	40.6 – 19.8j Ω	51.5 – 20.6j Ω

B. Optical Tracker Setup

An optical tracking solution is leveraged as the reference for measuring the stretching of the antenna sensor during the respiration cycle. The utilized optical tracking technology from Creaform (Levis, QC, CA) is a system consisting of C-Track™, an advanced dual-camera sensor system offering optical tracking and dynamic measurement, in conjunction with VXtrack™, a dynamic tracking software module capable of capturing the position and orientation of multiple objects in real-time. It provides highly accurate measurements with a volumetric accuracy of 0.068 mm in a 9.1 m³ scene [15].

The optical tracker operates by detecting specially designed marker stickers placed over the model as reference points, as shown in Fig. 3. In order to detect the stretching of the antenna sensor, three markers have been placed over the sensor at the ends of the antenna poles and the center feeding point. The volunteer was asked to wear the antenna sensor connected to the

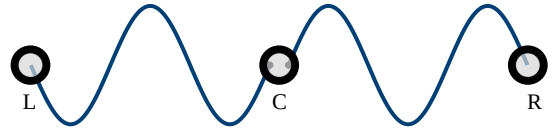


Fig. 3. Placement of the marker stickers on the sinusoidal antenna sensor. Labels are shown under each marker.

VNA system, stand in the tracker's field of view at a distance of 1.8 m, and lean against a wall to reduce the subject's movement.

The three-marker configuration makes it possible to calculate the amount of stretching for each pole of the antenna sensor. The 3D position of each marker along X, Y, and Z directions was tracked and recorded with a sampling frequency of 30 Hz and was exported as a CSV file. The marker points are named L, C, and R, from left to right for convenience. The overall setup diagram is shown in Fig. 4.

III. EXPERIMENTAL RESULTS

A. Data Preparation

The VNA setup and the optical tracking system were operated simultaneously, and the volunteer was asked to breathe normally. The data acquired from the VNA is the raw phase and magnitude of the antenna sensor's reflection, which is expected to fluctuate as strain is applied to the sensor [12]. To interpret the data from the optical tracker, the vector length between points L and C, and also R and C, is calculated as $l_L(t)$ and $l_R(t)$ respectively, to find the stretching of each pole of the antenna sensor relative to the center point throughout time.

$$l_L(t) = |\vec{r}_L(t) - \vec{r}_C(t)| \quad (1)$$

$$l_R(t) = |\vec{r}_R(t) - \vec{r}_C(t)| \quad (2)$$

Vector $\vec{r}(t)$ is the position vector of each marker point in space as measured relative to the tracker's origin throughout the time. After calculating (1) and (2), the baseline of each signal is

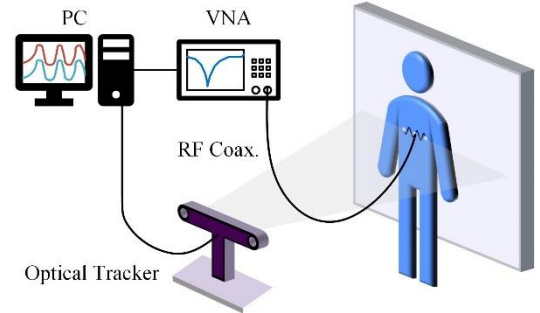


Fig. 4. The measurement setup. The volunteer is standing still and breathing normally while the sensor and optical tracker are operating simultaneously.

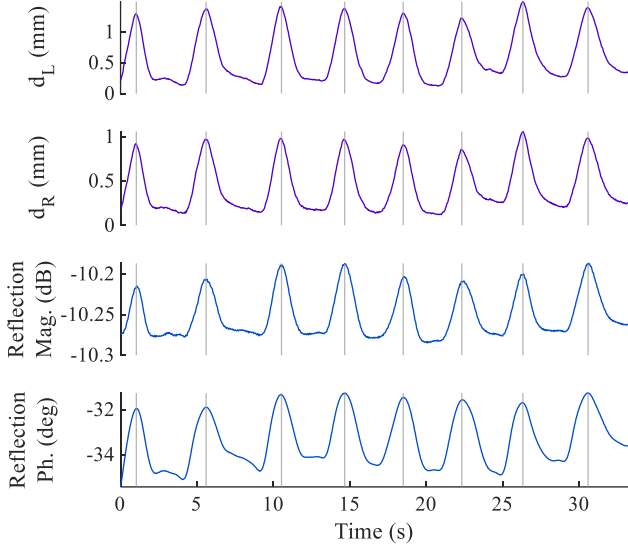


Fig. 5. The sample data acquire from the sensor and the reference. Gray lines indicate the peak timestamps from d_L reference signal.

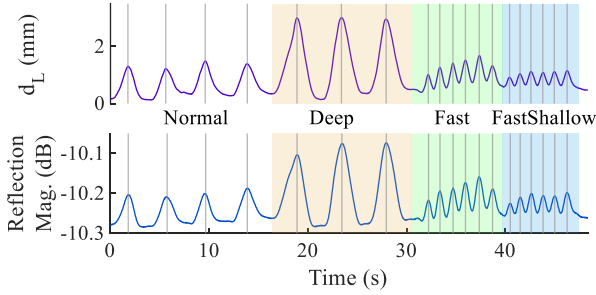


Fig. 6. Various respiratory behaviors.

subtracted to acquire the sensor stretching relative to the minimum length.

$$d_L(t) = l_L(t) - \min(l_L(t)) \quad (3)$$

$$d_R(t) = l_R(t) - \min(l_R(t)) \quad (4)$$

B. Normal Respiration

A set of signals acquired during a normal course of respiration is shown in Fig. 5. The presented curves include the stretching of the left and right poles of the antenna as seen through the optical tracker and the reflection magnitude and phase of the antenna sensor. The calculated mean square error (MSE) of the peaks acquired from the $d_L(t)$ and $d_R(t)$ signals are as low as 0.0015 s^2 , representing the high correlation between the stretching of the two antenna poles. To assess the error level against the worst case, an unusually high respiratory frequency of 100 breaths/min can be considered, equaling a period of 0.6 s, which is 400 times higher than the calculated MSE. This negligible error allows us to select one of the two as the primary reference against the antenna sensor measurements. Therefore, the measurements will be safely examined against $d_L(t)$ as the reference. The grey vertical lines shown in all plots indicate the timestamps of the peaks of respiratory activity waveform acquired from $d_L(t)$ reference.

Since the measured signal via the VNA setup is a complex value, both the Reflection Phase and the Reflection Magnitude can be utilized to extract the information. As shown in Fig. 5, both fluctuate during the respiratory activity. The calculated MSE for extracted peaks from d_L versus Reflection Magnitude and Reflection Phase is 0.0044 s^2 and 0.0023 s^2 , respectively, showing a significant correlation between the timings of the signal acquired through the antenna sensor and the reference.

C. Various Respiratory Behaviors

During the experiment, the volunteer was asked to breathe with different respiration depths and frequencies to analyze the capability of the antenna sensor to detect and differentiate between such conditions. The results are shown in Fig. 6, starting with normal breathing, then deep breathing, fast breathing, and finally fast and shallow breathing. As shown, the amount of the shift in Reflection Magnitude correlates with the respiration depth. This effect is not limited to the reflection magnitude; the same applies to the reflection phase. Similarly, an MSE is also calculated to assess the peak time accuracy in the respiration session with various respiratory behaviors, and the value of 0.0020 s^2 is acquired, reflecting the timing accuracy of the antenna sensor for different breathing profiles.

An extra assessment is also done to show the relative accuracy of cycle amplitudes. For this, both the sensor and optical tracker signals have undergone removal of the baseline and then normalized to their maximum to scale them in the $[0, 1]$ range. Afterward, the Mean Absolute Error (MAE) of the peaks amplitude of the normalized signals is calculated as 0.0395, implying a $\pm 3.95\%$ relative error in measured cycle amplitudes. These results suggest that using sinusoidal antenna sensors for respiratory monitoring can not only detect anomalies in respiratory frequency, potentially associated with Hyperventilation and Hypoventilation, but also can detect anomalies of breathing depths, which can be associated with Hyperpnea and Hypopnea.

D. Sleep Apnea Detection

In another experiment, the volunteer was asked to simulate a central apnea case by artificially emulating a cessation of breathing. Fig. 7 presents the signal where the volunteer mimics apnea amid a course of normal breathing. Two events are visible in the graph, where the first is a case of an end-inspiratory pause, and the second is an end-expiratory pause. Such an event can be detected by finding plateaus in the signal where the antenna's response is not changing higher than a threshold amount [16].

IV. CONCLUSION

An experimental setup consisting of a highly accurate optical tracker is employed to assess the accuracy of a wearable sinusoidal antenna sensor in detecting axial stretching due to the respiratory cycle. It is shown that the signal recorded using the antenna sensor is well correlated with the actual stretching detected by the optical tracker. The fluctuations in reflection magnitude and phase have shown a significant correlation with the reference signal with MSE values of 0.0044 s^2 and 0.0023 s^2 , respectively. Examples of various respiratory behaviors with different depths and frequencies were provided.

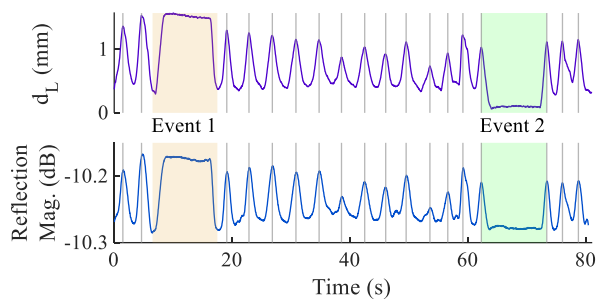


Fig. 7. A emulated case of Sleep Apnea. White background indicates normal respiration, Event 1 (yellow) indicates an end-inspiratory pause, and Event 2 (green) indicates an end-expiratory pause.

The robustness of this respiratory monitoring method is assessed for such cases from both aspects of timing accuracy, with an MSE of 0.0020 s^2 , and amplitude accuracy, with a relative error of $\pm 3.95\%$. The possibility of detecting frequency and depth anomalies potentially associated with hyperventilation, hypoventilation, hyperpnea, hypopnea, and sleep apnea cases has also been demonstrated.

Although the findings of this work suggest utilizing sinusoidal antenna sensors as a promising method for detecting respiratory cycles, some questions remain to be addressed in future studies. To ensure the reliability of this system as a respiration monitoring apparatus, the next step should be evaluating this device against a medical standard spirometer to understand the link between the actual airflow and the signal picked up by the sensor. Also, the tests can be extended to several volunteers from different ages, sexes, and physical builds to ensure the robustness and inclusivity of this method.

Moreover, this work was a short study limited to the frequency and cycle period accuracy via peak time analysis along with a cycle amplitude analysis; however, there are still more parameters to analyze, such as inspiration/expiration duration. Plus, future work can analyze all these parameters against the parameters acquired from the gold standard in an extensive Bland-Altman analysis for many participants with different physical specifications in various setups to demonstrate the accuracy and reliability of this method compared to a clinical reference device from a medical point of view [17].

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